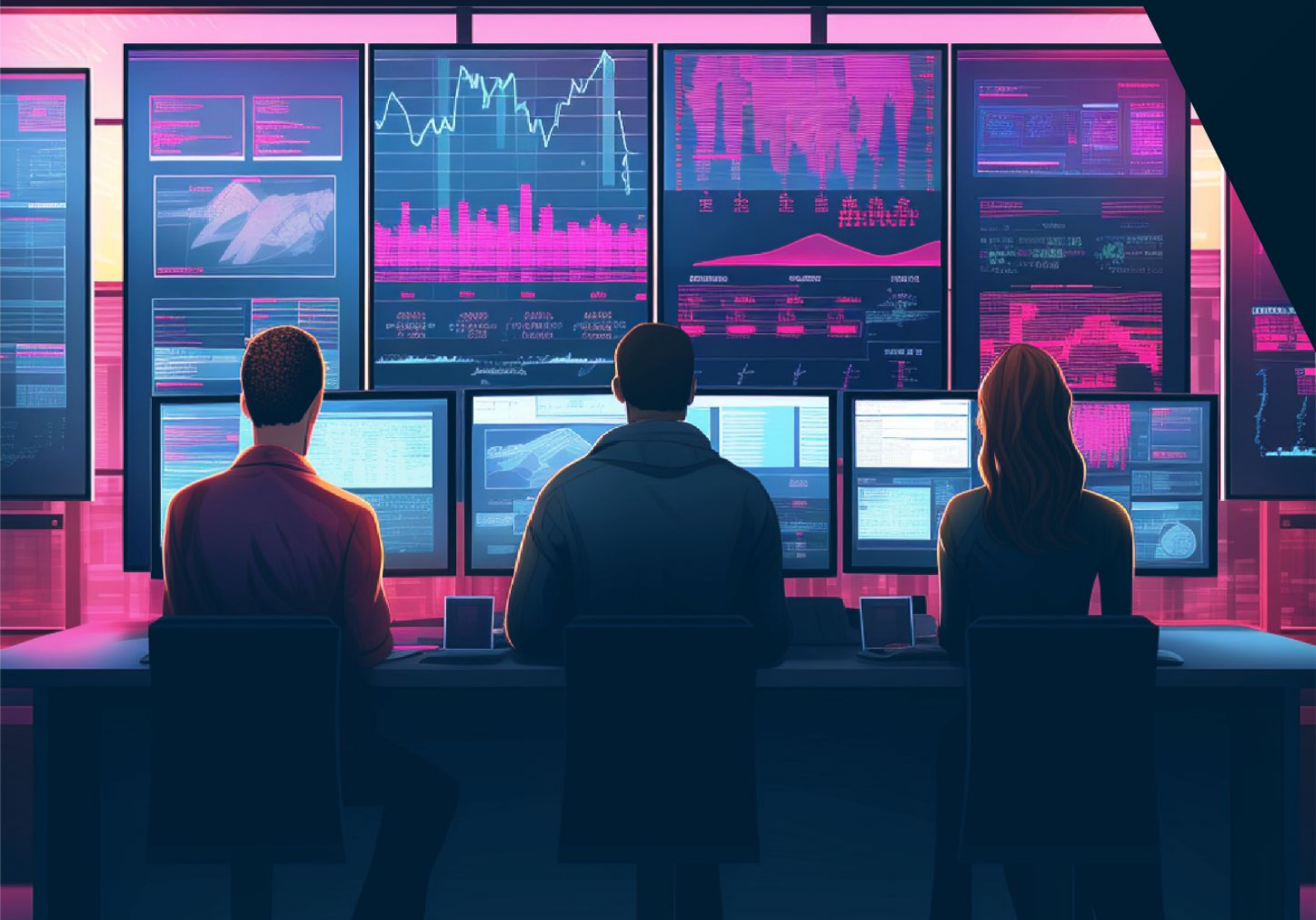




The Role of Breach and Attack Simulation in SOC Modernization

Discover how BAS enhances today's modern Security Operations Centers.



EXECUTIVE SUMMARY

A Security Operations Center (SOC) team comprises professionals who monitor, assess, and defend against cybersecurity threats and incidents. However, as neat as the theoretical definition sounds, the practical situation of modern SOC *does not look as promising*, given the constant battle against the expanding attack surface and evolving threat landscape.

The workload for SOC professionals is exceptionally high, with each expected to handle hundreds to thousands of daily alerts. This intense demand leads to burnout and frequently causes staff to leave their jobs, which *perpetuates a cycle of understaffing*. Because there are not enough team members, those who remain face overwhelming operational workloads, spending most of their time eliminating false positives. This monotonous task stifles their professional development and leads to frustration. Additionally, entry-level SOC members (Layer 1) often do not have enough time to thoroughly analyze alerts, so they must pass complex cases to more experienced staff (Layer 2 or 3) without a solid grasp of the underlying issues. This issue contributes to a widening skills gap within the organization. The constant need to train new employees and keep them on board drains resources further, exacerbating the problem.

This is why forward-thinking organizations invest in Breach and Attack Simulation (BAS) technologies. BAS outdoes traditional security assessment methods by *continuously* and *automatically* testing the existing defense measures against known and emerging threats. By automating routine tasks and providing *actionable mitigation suggestions*, BAS reduces staff workload, helping to mitigate burnout and potentially decreasing turnover rates. In addition, BAS serves as a practical training platform that enhances the skill sets of SOC teams by contextualizing the attack-kill chain of adversaries, as simulations are mapped to industry-level frameworks like MITRE ATT&CK, strengthening organizational defense capabilities and providing career development opportunities, which can help retain talent.

Finally, BAS enables organizations to make informed decisions about prioritizing security gaps. In a resource-limited environment, efficiently allocating efforts and focusing on the most critical security weaknesses is invaluable. Moreover, the actionable insights BAS provides help to optimize security strategies and resource allocation, leading to more *sustainable SOC operations*.

50%

SOC teams are performing understaffed.

Half of the SOC professionals think that their team is operating understaffed.

Voice of SOC 2023 Report

INTRODUCTION

Today's modern SOC's navigate a dynamic environment. They are like players in an intensely *fast-paced chess game*, where every move is critical, and the board changes continuously. New rules are introduced, and unfamiliar adversaries emerge at every turn, making the game increasingly challenging. The **Picus Red Report 2024** sheds light on this complexity, revealing the **333% increase** in defense evasive malware that hunts for and takes down organizations' security controls in a stealthy manner like a hunter-killer submarine [1]. These sophisticated changes in the threat landscape underscore the need for SOC's to adapt and enhance their tactical readiness constantly.

In this relentless game, there's a concerning gap between the perceived and actual effectiveness of SOC teams. A significant **66% of decision-makers** trust employees' abilities to counterattack cyber incidents more than they trust the capabilities of their security teams to identify and address security gaps (63%) and the accuracy of data alerts (59%) [2]. This perception starkly contrasts with the reality reported in the **Picus Blue Report 2023**: only 59% of attacks are effectively prevented, a mere 37% are logged, and just 16% trigger alerts. These figures highlight a pressing need for SOC's to reassess and improve their strategies [3].

BAS is a critical tool in modernizing SOC's, offering *continuous, automated, and realistic* simulations of sophisticated cyber attacks. This approach addresses the limitations of traditional security assessment methods and bolsters SOC's readiness for the ever-evolving threat landscape.

BAS tools also address the chronic issues of understaffing and burnout in SOC's, where 63% of practitioners face burnout, and 55% consider job changes annually [4]. Hence, BAS' automation capabilities are seen as a beacon of hope, potentially improving work-life balance for 93% of practitioners [4]. Integrating BAS reduces operational burdens by prioritizing critical issues and provides essential skill training, countering skill stagnation and enhancing team knowledge and proficiency in a dynamic, hands-on setting.

In conclusion, SOC's are pivotal in the modern world of cybersecurity. However, facing challenges of understaffing and burnout, these teams must embrace technologies like BAS to enhance readiness and skillfully address the continuously changing threat landscape.

66%

of decision makers trust employees' abilities to avoid cyber incidents more than the capabilities of security teams to identify and prioritize security gaps (63%), and the accuracy of data alerts (59%).

The State of Cyber Defense Report 2023

BREACH AND ATTACK SIMULATION (BAS)

The expanding attack surface due to digital transformation necessitates organizations to allocate more resources to their security infrastructure. However, merely investing in defensive technologies does not ensure the effective prevention and detection of cyberattacks. Thus, maintaining a robust security posture requires organizations to consistently test, validate, and promptly address any gaps in their security controls.

In this context, BAS has emerged as the most effective and efficient approach for evaluating security controls. Addressing the limitations that traditional security assessment approaches inherently have, BAS operates continuously, enabling organizations to gain real-time insight into the performance of their security controls and optimize them accordingly.

BAS further broadens its capabilities by simulating a diverse array of attack techniques used by threat actors, including malware, ransomware, APTs, web application and data exfiltration attacks, and exploiting vulnerabilities. These simulations replicate TTPs used by real adversaries, offering a realistic evaluation of an organization's defense mechanisms.

Use Cases of BAS for Security Operation Centers (SOC)

Security Posture Assessment

BAS tools evaluate an organization's security posture by simulating sophisticated cyber threats. By emulating known and emerging threat actors, they illustrate potential breach scenarios, assisting teams in readiness for real-world situations while avoiding the risks associated with traditional penetration testing.

Tuning of Security Tools

Through BAS exercises, SOC teams can harness detailed feedback to fine-tune critical tools such as firewalls, intrusion prevention systems, and antivirus software. This leads to enhanced detection and prevention, bolstering their overall security infrastructure.

Training and Awareness

BAS is vital in equipping SOC staff with skills to handle cyber threats, both emerging and known, in a controlled setup. These exercises test the team's detection and response capabilities and offer practical, hands-on learning, which is crucial for navigating the ever-changing cybersecurity landscape.

Continuous Security Validation

BAS enables SOC teams to validate their implemented security controls' effectiveness continuously. This ongoing validation is crucial because it allows SOC teams to identify misconfigured or non-functioning security solutions, ensuring they operate as intended.

Reducing the Risk of Breaches

BAS simplifies the identification of critical vulnerabilities in security postures and provides actionable mitigation strategies. It produces detailed reports on potential cyber attack scenarios and prioritizes actions to tackle the most significant business risks effectively.

Detection and Response Testing

BAS allows SOC teams to rigorously test detection and response mechanisms, including SIEM rules and EDR systems, in simulated attack scenarios. This approach facilitates fine-tuning existing and newly implemented rules and enhances incident response processes, ensuring a seamless and effective security response to potential threats.

Cost-Effective Security Assessment

BAS offers a cost-effective alternative to traditional security testing methods like red teaming and penetration testing. These conventional approaches, while valuable, often provide only a snapshot of an ever-changing security infrastructure and can be resource-intensive. Through its continuous and automated practices, BAS not only reduces labor and associated costs but also ensures a more dynamic and up-to-date evaluation of security postures.

Red and Blue Team Augmentation

BAS technologies significantly enhance the capabilities of red teams (specializing in offensive security strategies) and blue teams (focusing on defensive security measures). By automating routine tasks, these simulations free up these teams to concentrate on developing and countering more sophisticated and specialized attack scenarios.

ENHANCING SOC EFFICIENCY WITH BAS

This section highlights five key ways in which BAS improves the efficiency of modern SOC's.

1. Elevating Threat Readiness with BAS

The threat landscape is rapidly evolving, marked by the increased volume and sophistication of attacks. Attackers are using advanced and stealthy tactics, techniques, and procedures (TTPs) and becoming more adept at evading traditional defensive measures. This is also proven by [Picus Red Report 2024](#), which highlighted the 333% increase in the use of defense-evasive malware that *hunts down and kills* organizations' implemented defensive controls in a stealthy manner.

Furthermore, the challenge for SOC teams is compounded by the rise of state-sponsored activities and APT groups, which can remain undetected in networks for extended periods. As a result, SOC teams find themselves in a position where they must continuously adapt their threat detection and response strategies to address known and emerging threats effectively. Unfortunately, keeping pace with these advanced threat actors is becoming increasingly challenging for organizations of all sizes as they struggle to adjust to the rapidly evolving environment and accurately assess their threat readiness. This constant pressure to adapt is further exacerbated when high-profile cyber threats emerge, requiring immediate and intense attention from executive management.

For instance, when a high-profile cyber threat arises, executive management's intense pressure triggers a flurry of urgent inquiries regarding the threat's specifics, the SOC's analytical methods, and the effectiveness of its detection, prevention, and incident response systems. This pressure disrupts routine tasks, creating a tense and potentially chaotic environment within the SOC as its members scramble to meet these demanding expectations.

Fortunately, BAS tools play a pivotal role in addressing the increasing complexity of cyber threats and the challenges faced by SOC teams in staying ahead of sophisticated adversaries. It systematically exposes an organization's defenses to a wide range of simulated attack vectors in a controlled, non-destructive environment. This method allows for the identification of weaknesses in the security posture, enabling swift remediation before these gaps can be exploited by adversaries.

Here are some of the attack vectors that organizations can simulate with BAS.

✓ **Vulnerability Exploitation Attacks**

Through BAS, organizations can simulate attacks exploiting known software vulnerabilities and zero-day threats in operating systems, applications, and services on individual hosts, as well as target vulnerabilities in network-level systems. Utilizing publicly available proof-of-concepts, these simulations rigorously test the effectiveness of both host-level security solutions and network-level defenses.

✓ **Malware Attacks**

BAS is capable of conducting controlled malware simulations across a range of malware types, verifying that prevention, detection, and quarantine mechanisms function optimally to mitigate risks associated with malicious software.

✓ **Lateral Movement Attacks**

Once a host is compromised, attackers often attempt lateral movements within a network to access other systems and data. BAS can simulate techniques like pass-the-hash that test the ability of the organization's network to contain and stop adversaries from moving laterally.

✓ **Endpoint Attacks**

BAS can simulate endpoint attacks on various operating systems, including Linux, macOS, and Windows. This allows organizations to assess the resilience of their endpoint security solutions across different platforms and environments.

✓ **Phishing Emails**

SOC teams can effectively test their organization's email security solutions by leveraging BAS to simulate sophisticated phishing attacks. This allows them to thoroughly evaluate their capability to detect and robustly quarantine malicious emails containing malware links.

✓ **Data Exfiltration**

Simulating data leakage attempts allows organizations to validate the effectiveness of their Host-based Data Loss Prevention (HDLP) solutions and network monitoring practices. Such testing ensures that sensitive information remains secure, even in the event of perimeter defenses being breached.

✓ **Post-exploitation Behaviors**

Simulating post-exploitation activities is crucial for assessing the detection and response capabilities of security systems following an initial breach. BAS can simulate a range of activities that attackers typically engage in after gaining access to a system, such as establishing persistence, escalating privileges, installing backdoors, or disabling security software.

✓ **Advanced Persistent Threat Campaigns**

BAS is proficient in simulating multi-vector attack chains, providing a comprehensive method to evaluate an organization's preparedness against sophisticated adversaries. This functionality involves replicating the complex, sustained, and stealthy tactics typical of APTs.

2. Enhancing Skill Development

The 2023 SANS Institute SOC Survey reveals a marked **shortage of skilled staff**, a concern that becomes more significant when considering the increasing complexity of cyber threats [5]. These evolving threats demand an expertise level that is not readily abundant in the current cybersecurity workforce, emphasizing the pressing requirement for personnel capable of effectively managing and mitigating these advanced risks. Consequently, the need for training and retaining SOC personnel is universally acknowledged.

Typically, companies resort to hiring third-party organizations for such training needs. However, these training programs often fall short as they tend to be lengthy, requiring SOC operations to be put on hold, and they are *not specifically tailored to the organization's unique playbook* or the specific *security controls* used within the company. This limitation means that SOC analysts can only be trained up to a certain level, leaving an advanced knowledge gap that *third-party training and even certification processes fail to fill*.

✓ In-depth Training on Cyberattack Techniques and APT Groups Using BAS

BAS tools provide efficient and tailored in-house training aligned with the organization's unique playbook. Through simulating advanced attack patterns and exposing analysts to TTPs of advanced APT groups (each mapped to frameworks like MITRE ATT&CK), analysts develop a comprehensive understanding of cyber kill chains employed by real adversaries. This enables SOC teams to remove the defender bias, gaining insights from an attacker's viewpoint. Such realistic experiences enhance their ability to comprehend potential security breaches and improve their alert management proficiency.

Furthermore, BAS simulations allow analysts to expand their skill sets, familiarizing them with even, for instance, the latest evasion and persistence techniques. This is mainly due to BAS platforms' regularly updated threat libraries, which keep SOC teams current with the latest cyber threat intelligence. Consequently, they better identify attackers' strategies and effectively predict future attack vectors. Ultimately, BAS tools effectively address skill stagnation, which is often a leading cause of high turnover rates in the field. This comprehensive approach to training ensures that SOC teams are always equipped with the most current and relevant skills needed.

15% of SOC practitioners are pursuing advanced certifications, 9% are focusing on how to operationalize MITRE ATT&CK, and 7% are engaged in learning penetration testing to combat skill stagnation.

The Voice of the SOC 2023

3. Decreasing Operational Burnout

The 2023 RSA Conference Survey conducted by Gurukul illuminates the operational burden experienced by SOC teams [6], which contributes to increasing turnover rates. The survey reports that SOC teams are often overwhelmed by the sheer volume of security alerts, with some receiving over 1,000 alerts daily. This inundation underscores the need for skilled professionals to process and prioritize these alerts efficiently.

Moreover, the 2023 SANS SOC Survey report casts a concerning spotlight on employee burnout in SOC teams, revealing an alarming pattern of high turnover rates. Notably, only 38% of SOC employees stay in their roles for one to three years, with nearly 6% leaving in less than a year [5]. Further amplifying these concerns, the Voice of SOC 2023 report indicates that 63% of practitioners experience burnout, and 55% consider job changes within the year [4].

✓ Improving Alert Management Processes

By utilizing BAS, SOC analysts actively participate in simulated real-life attack scenarios, enhancing their skills in prioritizing and managing security alerts. When an attack simulation is executed, it triggers an alert within the organization's SIEM system, replicating, for instance, malicious network traffic closely resembling a real cyberattack. The analyst's responsibility is to quickly evaluate and prioritize the alert, assessing its severity and potential impact on the business operations. This process entails conducting a thorough analysis of the alert, followed by the development of a strategic response.

✓ Automating Detection Content Analysis and Development

BAS technology provides SOC analysts with a specialized approach for developing and refining signatures and rules. By deploying BAS-driven attack simulations, including web and email-based threats, analysts can validate if newly implemented signatures or rules trigger alerts within their SIEM system. This essential process ensures the accuracy and effectiveness of these security measures. The continual rerunning of BAS simulations allows for rigorous testing and validation of new signatures against various simulated cyber threats, ensuring they are robust and functional in real-world scenarios, thereby fortifying the network's defense.

SOCs are also grappling with staffing issues, with 59% of leaders indicating their teams are understaffed.

*State of Cybersecurity 2023
By ISACA*

4. Having Swift and Robust Response to Threats

The evolving threat landscape and the potential for revenue and reputational damage have made a strong security posture crucial for organizations. This often leads to significant investments in security controls. However, a common error for organizations is to assume that their implemented defensive measures function as intended. Many organizations mistakenly believe their security controls are properly configured and fully effective against cyber threats. This assumption can be dangerously misleading as there is a widespread issue with misconfigured, disconnected, turned-off, or non-optimized security tools.

The [Blue Report 2023](#) by Picus underscores this challenge with concerning statistics: only 59% of attacks are prevented, indicating a significant gap in the ability of SOCs to block cyber threats effectively. More alarmingly, only 37% of attacks are logged after infiltrating environments and only 16% trigger alerts. This low alert rate impedes security teams' ability to identify and respond promptly to potential threats, leaving organizations vulnerable to sophisticated cyberattacks. These statistics reveal a troubling reality: many organizations are operating under a *"false sense of security"*, unaware of the inadequacy of their existing controls in detecting and responding to advanced cyber threats.

BAS effectively addresses the challenge of inadequate threat response by providing a comprehensive and realistic evaluation of an organization's security posture. It uncovers non-functioning security controls and identifies misconfigurations that hinder effective threat response, allowing for immediate remediation. Below are the key solution points that BAS technologies contribute to effective threat response.

✓ Automated Security Control Validation

BAS systematically validates security controls across all layers of an organization's defense-in-depth strategy. This validation ranges from the outermost network layer to the application, host, and innermost data layers. It conducts diverse attack simulations at each layer and evaluates the effectiveness of cross-layer security solutions.

✓ Enhanced Threat Intelligence Integration

BAS not only validates existing security controls but also significantly improves an organization's threat intelligence capabilities. By simulating current and emerging cyber threats, BAS offers a dynamic and updated perspective on the threat landscape. This integration allows for the continuous assimilation of new threat data, ensuring that SOC teams are not just relying on historical data but are also prepared for novel attack vectors.

✓ Actionable Mitigation Strategies and Suggestions

BAS solutions extend beyond merely identifying coverage gaps; it empowers organizations with robust tools to address these gaps decisively. BAS platform presents an extensive array of mitigation suggestions, encompassing both vendor-specific and generic solutions such as signatures and detection rules. This enables a highly customized response to a wide spectrum of security challenges.

5. Improved Risk Prioritization and Resource Allocation

Modern SOC teams are confronted with the complex challenge of prioritizing vulnerabilities in their security posture, compounded by frequent misconfigurations, non-optimizations, and discontinuations in security products. The rapidly evolving nature of threats, including sophisticated malware, ransomware, and APTs, alongside the emergence of zero-day exploits and novel attack techniques, adds to the complexity, making the threat landscape increasingly unpredictable and difficult to navigate.

For modern SOC teams, especially those that are understaffed, pinpointing the most critical vulnerabilities to business operations is a task marked by significant uncertainty. This challenge is magnified by the constantly shifting threat environment, where the varied risk levels and potential impacts of threats make accurately assessing vulnerabilities in the security posture even more difficult. In such a dynamic landscape, the consequences of overlooking or underestimating a vulnerability are severe, potentially jeopardizing an organization's operational security and overall stability. This highlights the critical and complex nature of risk prioritization that modern SOC teams must contend with.

BAS allows SOC teams to streamline their risk evaluation and concentrate their resources on the most critical weaknesses for the business.

✓ Streamlined Risk Evaluation

BAS tools elevate risk evaluation by simulating realistic cyber-attack scenarios, enabling SOC teams to identify and prioritize weaknesses that are most likely to be exploited. This refined focus helps direct security efforts effectively, ensuring the most critical risks are addressed first.

✓ Adaptability to Threat Landscape

BAS solutions constantly adapt to new threats and attacker tactics, providing up-to-date insights into the organization's security posture. This adaptability allows SOC teams to refine their risk prioritization strategies continually, staying ahead of evolving cyber threats and maintaining a proactive defense approach.

✓ More Focused Resource Allocation

BAS tools identify realistic attack vectors, enabling a strategic and data-driven approach to resource allocation. By avoiding the overload of non-prioritized issues common in traditional methods, BAS ensures that limited security resources are concentrated on areas with the highest potential impact.

✓ Strategic Business-Centric Prioritization

BAS assists SOC teams in aligning defense prioritization with the organization's business objectives. By evaluating the criticality of different systems to business operations, BAS guides security efforts to focus on protecting the most vital components of the organization's infrastructure.

6. Effective Collaboration Between SOC and Management

Effectively communicating the SOC team's needs to stakeholders presents a significant challenge for SOC leaders and managers. More than just securing budget approvals, their role involves clearly explaining the SOC's specific needs and current status within the organization's broader financial decision-making structure. This becomes particularly intricate in sectors less regulated than industries like banking, where cybersecurity regulations are more stringent and compliance is viewed as mandatory. In less regulated fields, compliance is often seen as optional, making it essential for companies to independently develop a strong cybersecurity awareness.

In such environments, the involvement of upper management in understanding and supporting cybersecurity needs is vital. SOC leaders must provide an unbiased and clear presentation of the current security posture and the necessary improvements. While engaging external consultancy services for an independent assessment might be considered, the high costs associated with these services can cause hesitation among stakeholders. This further complicates the process of making informed decisions about cybersecurity investments and strategies.

BAS provides clear, data-driven insights into cybersecurity risks and defenses, aiding in strategic communication with stakeholders and facilitating informed decision-making for resource allocation and cybersecurity policy development.

✓ Risk Identification and Assessment

BAS tools are essential in identifying and assessing security risks. They simulate various attacks to test an organization's defenses, highlighting vulnerabilities and effectiveness of existing security measures. This process is crucial for conveying security needs to stakeholders, especially when requesting additional resources or strategic changes.

✓ Strategic Insight and Planning

BAS provides valuable insights into security readiness. Continuous simulations by SOC's evaluate defenses against evolving threats. This ongoing assessment is vital for strategic cybersecurity planning, offering insights into strengths and weaknesses and guiding the development of proactive strategies.

✓ Data-Driven Communication

Using BAS, SOC's adopt a data-driven approach in communicating with stakeholders. Simulations yield measurable results on defense capabilities, effectively conveying the state of security and its potential business impacts. This objective data informs decision-making, emphasizing the urgency of cybersecurity investments.

✓ Stakeholder Engagement and Education

BAS plays a critical role in engaging and educating stakeholders about cybersecurity challenges. By illustrating the consequences of specific threats, it makes cybersecurity issues tangible and understandable. This approach ensures stakeholders perceive cybersecurity as a fundamental aspect of organizational health and resilience, not merely an IT issue.

CONCLUSION

In the increasingly sophisticated threat landscape, the introduction of BAS into SOC is emerging as a pivotal game-changer. BAS transcends traditional reactive measures, embedding a proactive and dynamic approach into the core of cybersecurity operations. The adoption of BAS addresses the multifaceted challenges SOC face in this era of digital complexity, serving as an invaluable tool for realistic threat simulation. This enables continuous validation and enhancement of security measures, which is critical given the rapid evolution of cyber threats that demand not just robust responses but forward-thinking anticipation.

Furthermore, BAS significantly reduces the operational burden on SOC personnel, directly tackling issues such as burnout and skill stagnation, thereby fostering a more sustainable working environment. Strategically, BAS provides a decisive advantage in prioritizing security gaps and vulnerabilities. Its data-driven insights offer a clearer perspective on resource allocation, enhancing the overall efficiency and effectiveness of cybersecurity strategies. Moreover, BAS empowers SOC teams to communicate their needs and challenges more effectively to stakeholders, ensuring that cybersecurity is recognized as a critical and integral component of organizational resilience and success.

The Verdict:

Implementing BAS in SOC operations is more than just an upgrade; it's a significant evolution in cybersecurity strategy. This integration marks a transition from a reactive stance against cyber threats to establishing proactive control and foresight. BAS elevates SOC from passive defenders to active guardians of digital integrity, equipped to not only confront but also outpace evolving cyber threats.

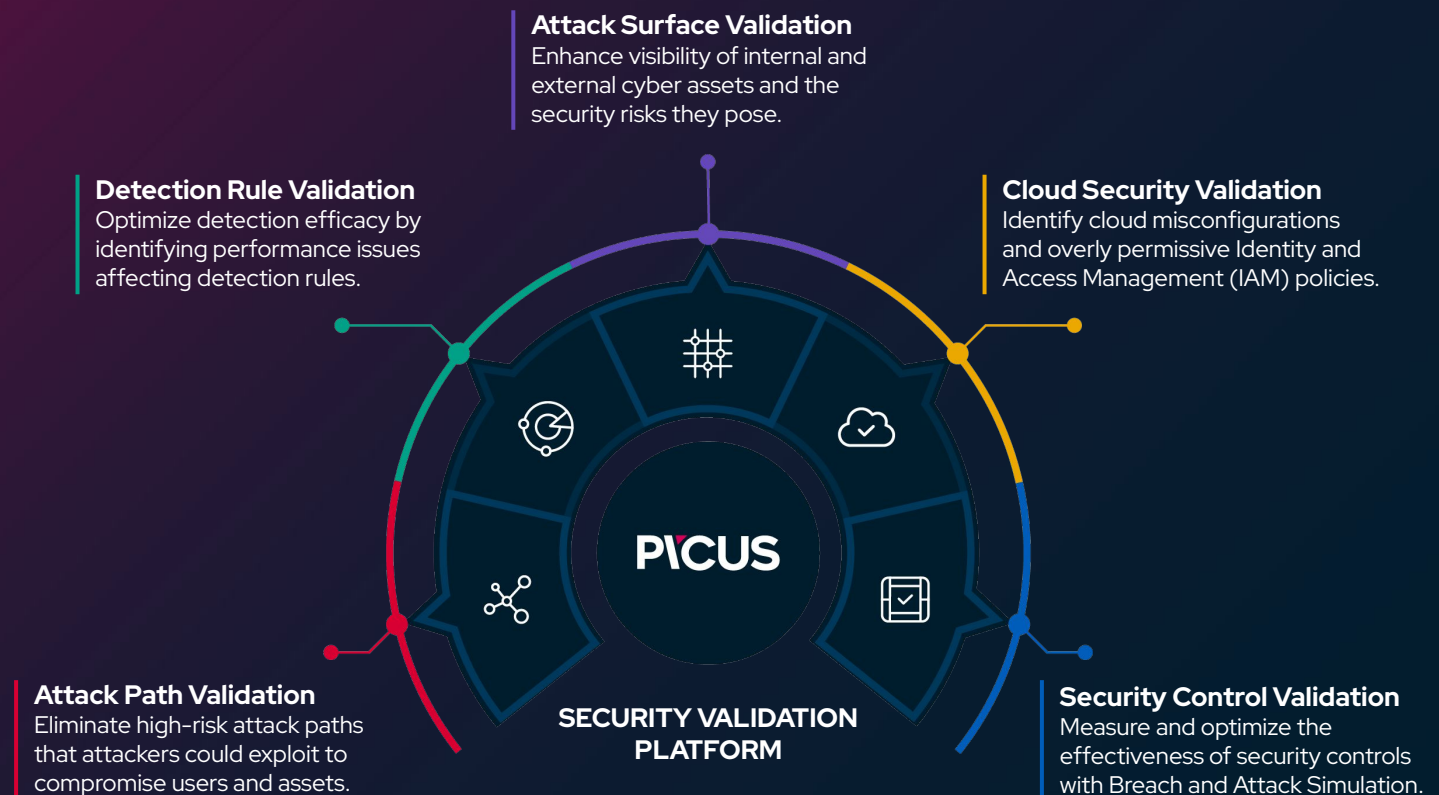
This strategic enhancement is not simply a response to current challenges but a proactive measure for future-proofing cybersecurity defenses. By adopting BAS, organizations are not merely defending against today's threats; they are preparing for tomorrow's unknowns. This shift redefines resilience, enhances operational effectiveness, and transforms cybersecurity from a defensive requirement into a strategic asset. In an era where cyber threats are a constant reality, BAS serves as a beacon of innovation and readiness, steering SOC towards a more secure and confident digital future.

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ABOUT PICUS

As the pioneer of Breach and Attack Simulation, Picus Security helps security teams consistently and accurately validate their security posture. Our **Security Validation Platform** simulates real-world threats to evaluate the effectiveness of security controls, identify high-risk attack paths to critical assets, and optimize threat prevention and detection capabilities by including:



**Elevate your security capabilities with the
Picus Security Validation Platform**

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